

MLOPS IN PRODUCTION · ISSUE 03 OF 08

Data drift vs concept drift.

Drift is a question, not an action.

03

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Two different problems

DATA DRIFT

- Inputs change: $P(X)$
- Detectable without labels
- New users, new channel
- Model may still be fine

CONCEPT DRIFT

- Relationship changes: $P(y|X)$
- Needs labels or proxies
- Fraudsters adapt
- Model is now wrong

How to detect it

METHOD	USE FOR	WATCH OUT
PSI	Binned features	Bin choice matters
KS test	Continuous features	Too sensitive at large n
Prediction distribution	Early proxy	Not proof of harm
Labeled sample accuracy	Concept drift	Label delay

When an alert fires

- 1 Detect
- 2 Check for a data bug
- 3 Real or seasonal?
- 4 Ignore, retrain or roll back
- 5 Eval gate
- 6 Promote

Most “drift” I have seen was a broken join or a changed default value upstream.

Alerting on drift

DO

- ✓ Weight features by importance
- ✓ Compare to a seasonal baseline
- ✓ Alert on sustained shift

AVOID

- ✗ Alert on every p-value
- ✗ Auto-retrain on any shift
- ✗ Ignore known seasonality

Examples

01 Data drift

A new campaign brings a younger audience.

02 Concept drift

A policy change alters what “churn” means.

03 Not drift

An upstream null became a zero.

KEY TAKEAWAY

Before you retrain, rule out a data bug.

- Data drift: inputs move
- Concept drift: the relationship moves
- Triage first, act second

YOUR TURN

Was your last drift alert real drift or a pipeline bug?

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